



Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2
3  void func(int n) {
4      int count = 0;
5      if (n == 1) {
6          count++; // for the execution of if condition
7          printf("*");
8      }
9      else {
10         count++; // for the execution of if condition
11         count++; // for i = 1 initialization
12         for (int i = 1; i <= n ; i++) {
13             count++; // for the successful true condition checks
14             count++; // for j = 1 initialization
15             for (int j = 1; j <= n ; j++) {
16                 count++; // for the number of iterations of the inner loop
17                 count++; // for the break statement
18                 break;
19             }
20             count++; // for the outer loop increment (i++)
21         }
22     }
23     printf("%d", count);
24 }
25
26 int main() {
27     int n;
28     scanf("%d", &n);
29     func(n);
30     return 0;
31 }
32
```

Check

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓
✓	143	717	717	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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